MILITARY SPECIFICATION

CARTRIDGE, ILLUMINATING, M301A2 WITH FUZE, TIME, M84 FOR 81MM MORTARS LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of illuminating cartridge, designated as Cartridge, Illuminating, M301A2, W/Fuze, Time, M84.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of
MIL-P-46917 - Primer, Percussion, M54, Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1315

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.

DRAWINGS

ORDNANCE CORPS

7548648 - Container, Ammunition, Metal, M266 for Cartridge, Illuminating, M301A2 for 81MM Mortars. Assembly, Details, Packing and Marking
7548649 - Box, Packing, Ammunition for 81MM Mortar Ammunition in Metal Container M266. Assembly, Details and Packing
8865058 - Cartridge, Illuminating, 81MM, M301A2, with Fuze, Time, M84

PUBLICATIONS

ORDNANCE CORPS

IEL-8865058 - Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Material and parts shall be in accordance with applicable specifications and drawings.

3.2 Loaded cartridge.-The loaded cartridge shall comply with all requirements specified on Drawing (dwg.) 8865058, and with all requirements specified in applicable specifications.

3.3 Disassembly torque

3.3.1 Fuze.-The fuze shall withstand a disassembly torque of 200 inch pounds when tested as specified in 4.3.1.

3.3.2 Fin assembly.-The fin assembly shall withstand a disassembly torque of 300 inch pounds when tested as specified in 4.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.

3.4 Functioning (proving ground).-The cartridge shall comply with the following requirements when tested as specified in 4.3.2.

3.4.1 Firing.-The cartridge shall not fail to fire.

3.4.2 Fuze.-The fuze shall function and cause separation of the body tube assembly and the tail cone.

3.4.3 Burning of illuminant.-The illuminant shall ignite and burn for not less than 60 seconds.

3.4.4 Operation of parachute.-The parachute shall open after separation of the body tube assembly and tail cone and suspend the illuminant assembly at a rate which will assure effective illumination (see 6.6) of the illuminant for the duration specified in paragraph 3.4.3.

3.4.5 Premature burst.-There shall be no premature burst or metal parts failure in the barrel of the mortar or within 200 yards horizontal range of the mortar.

3.4.6 Horizontal range.-No cartridge shall have a horizontal range less than 80 percent of the average horizontal range.

3.5 Workmanship

3.5.1 General.-All manufacturing, processing and assembly operations shall be correctly performed in a thorough, workmanlike manner. Parts shall be free of burrs, chips, dirt, grease, corrosive material and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

3.5.2. Painting and marking.-The painting and marking shall be restored by the loading and assembling contractor on any components furnished to him painted or marked when such painting or marking has been marred or obliterated.

Source: https://assist.dla.mil - Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspection performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:1
Check the source to verify that this is the current version before

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions

4.2.1 Lot formation.-A lot shall consist of loaded cartridges produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting safety, performance, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. Lots shall be submitted for inspection in accordance with Standard MIL-STD-105 (or Standard MIL-STD-1235, when applicable). Each lot shall contain:

a. Loaded projectiles from one interfix lot number from one manufacturer.

b. Loaded ignition cartridges from one interfix lot number from one manufacturer.

c. Loaded time fuzes from one interfix lot number from one manufacturer.

d. Loaded primers from one interfix lot number from one manufacturer.

e. Propellant increments made from not more than one lot of propellant from one manufacturer.

f. Fin assemblies from one interfix lot number from one manufacturer.

Source: https://assist.dla.mil Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.

4.2.2 Examination.-Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.5.

4.2.2.1 Cartridge, prior to assembly of fuze and fin (see dwg. 8865058)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.50 percent		
201.	Threads damaged	Visual	01001
202.	Sealing compound missing (where required)	Visual	01002
203.	Evidence of poor workmanship (see 3.5)	Visual	01003

4.2.2.2 Cartridge assembly (see dwg. 8865058)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Fuze not set safe	Visual	02001
2.	Safety pin improperly engaged .	Visual	02002

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before us

3.	Primer missing or above flush .	Visual	02003
4.	Propellant increment missing ..	Visual	02004
5.	Fin broken or distorted	Visual	02005
Major:	AQL 0.65 percent		
101.	Ignition cartridge missing or inverted	Visual	02006
102.	Total length, max.	Gage	02007
103.	Torque test failure of fuze (see 4.3.1)	Test	02008
104.	Torque test failure of fin assembly (see 4.3.1)	Test	02009
105.	Any component damaged or incorrectly assembled (except as described by Major defect 101 above)	Visual	02010
Minor:	AQL 1.50 percent		
201.	Protective coating with total damaged area in excess of 1/4 square inch	Visual	02011
202.	Marking misleading or unidentifiable, including incorrect color code	Visual	02012
203.	Evidence of poor workmanship (see 3.5)	Visual	02013

4.2.2.3 Unsealed metal container (see dwg. 7548648)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Spacer disc assembly missing ..	Visual	03001
Major:	AQL 0.40 percent		
101.	Support missing	Visual	03002
102.	Propellant increment cut or torn	Visual	03003

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:172
Check the source to verify that this is the current version before u

Minor:	AQL 1.50 percent		
201.	Glue on round	Visual	03004
202.	Cartridge improperly assembled into container	Visual	03005
203.	Round cannot be removed by hand	Manual	03006

4.2.2.4 Sealed metal container (see dwg. 7548648)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.50 percent		
201.	Contents loose	Manual	04001
202.	Marking misleading or unidentifiable	Visual	04002
203.	Protective coating damaged	Visual	04003

4.2.2.5 Sealed wood packing box (see dwg. 7468649)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Board broken or split	Visual	05001
102.	Strapping missing, broken or loose	Visual	05002
103.	DOD symbol missing or incorrect	Visual	05003
Minor:	AQL 1.50 percent		
201.	Contents loose	Manual	05004
202.	Strapping improperly assembled.	Visual	05005
203.	Marking misleading or unidentifiable	Visual	05006
204.	Carseal missing, unsealed or improperly positioned	Visual-Manual	05007

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.

4.2.3 Testing

4.2.3.1 Check test for possible deterioration of Primer, Percussion, M34 (see Specification MIL-P-46917 (MU). If the total elapsed time between original acceptance of the primer lot and the assembly of that lot into the loaded cartridges exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-P-46917 (MU) immediately before the primer lot is assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the loaded cartridges (see 6.7).

4.2.3.2 Profile and alignment (see dwg. 8865058) - Major defect - Code No. 02014.-The shell shall be gaged 100 percent for profile and alignment. Any shell which fails to comply with the requirement specified shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.5.

4.2.3.3 Proving ground.-The cartridges shall be observed for compliance with the requirements as classified in Table I.

TABLE I

Defect	Defect Classification	Code No.
Premature burst or parts failure at less than 200 yards horizontal range (see 3.4.5)	Critical	02014
Cartridge fails to fire (see 3.4.1)	Major	02015
Fuze fails to function and separate body tube assembly and tail cone (see 3.4.2)	Major	02016
Parachute fails to open and suspend illuminant at proper rate of descent (see 3.4.4)	Major	02017
Illuminant fails to ignite or burns less than 40 seconds (see 3.4.3)	Major	02018
Range less than 80 percent of average horizontal range (see 3.4.6)	Major	02019
Illuminant burns at least 40 seconds but less than 60 seconds (see 3.4.3)	Minor	02020

Source: https://assist.dla.mil - Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.

4.2.3.1.1 For the first lot produced and continuing until three consecutive lots have complied with the requirements specified herein a sample of eighty (80) randomly selected loaded assemblies from each lot shall be submitted for proving ground tests. If a total of six (6) or more major defects, or eight (8) or more minor defects, or one (1) or more critical defects occur during the test (see Table I), the lot shall be rejected. The test shall be as specified in 4.3.1.

4.2.3.1.2 After three consecutive lots have met the criteria of 4.2.3.1.1, the sampling plan for this test shall be in accordance with Standard MIL-STD-105, Code letter G (single sampling), utilizing an AQL of 2.5 percent for Major defects and an AQL of 4.0 percent for Minor defects (see Table I). The lot shall be rejected if a Critical defect occurs during any portion of the test or if the number of defectives equals or exceeds the rejection number specified. The test shall be performed as specified in 4.3.1.

4.2.4 Air pressure test of container (see dwg. 7548648) - Major defect - Code No. 04004.-This test shall be performed 100 percent. If any container fails to meet the requirement specified, it shall be classed defective and removed from the lot. The test shall be performed as specified in 4.3.3.

4.2.5 Inspection Equipment.-Index of Inspection Equipment Number IEL-8865058 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.5.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17
Check the source to verify that this is the current version before

4.2.5.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.2.5.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures

4.3.1 Torque test.-The disassembly torque shall be applied to the applicable component while holding the adjacent mating component in such a manner as to avoid possible loosening or damage to any component of the assembly.

4.3.2 Proving ground.-The test shall be performed at a Government proving ground in accordance with the applicable Ordnance Proof Manual and Acceptance Test Procedure. The cartridge shall be fired from an 81MM mortar, for which the round is standard, with the elevation set at 53 to 55 degrees. Four propellant increments shall be used on each cartridge for this test. Observation shall be made for compliance with the requirements specified and, in addition, delay time and burning time shall be recorded.

4.3.3 Air pressure test of container.-The sealed metal container shall be subjected to an external gage pressure of 3 p.s.i. Any container which shows a decrease in pressure during a 15 second interval shall be classed defective. The test equipment shall be as specified in paragraph 4.2.5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The cartridges shall be preserved and packaged in accordance with dwg. 7548648.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use

5.2 Packing

5.2.1 Level A.-The cartridges shall be packed in accordance with dwg. 7548647.

5.3 Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify title, number and date of this specification.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.

CDAF - Contractor's design responsibility on Army format in accordance with MIL-D-45608.
CDCF - Contractor's design responsibility on contractor format.
AD - Army design.
ADMU - Army design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control drawing.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawing.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black line on white background and shall provide a legible print produced via the 35MM microfilm process. Contractor shall submit one copy of the final design as a rolled set or flat set.

6.3.7 Submission of contractor designs.-Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Submission of initial production samples.-Instructions as to the location for evaluation of initial production shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.

6.5 Effective illumination.-Effective illumination is defined as burning of the canister while being suspended by the parachute. Burning of the canister after striking the ground shall not be considered as effective illumination. Any occurrences which detract from effective illumination as herein defined shall be recorded on the firing record.

6.6 Cost of check test.-The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project No. 1315-A423

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:17Z
Check the source to verify that this is the current version before use.